

AI Engineer with **4+ years of AI/ML experience**, including **3+ years building production systems**. Specialized in production LLM, RAG and agentic AI, with experience spanning retrieval, model serving, inference optimization, orchestration, cloud deployment and evaluation. Delivered **3× LLM inference throughput at 47% lower cost**, built multi-agent systems for finance, and currently develop AI accelerators within ALTEN's **A3**.

Experience

AI Engineer — A³ (AI at ALTEN)

ALTEN Group — AI for Management System & Accelerators

June 2026 – Present

Rabat, Morocco

- Built and productionized an **AI references-discovery platform** in 3 months (React, FastAPI, PostgreSQL/pgvector, Azure) for ALTEN's sales teams and directors; **101 active users** and **1,103 searches** in early rollout.
- Built the RAG pipeline (Mistral OCR and embeddings, OpenAI fallback) with an **LLM reranker** that selects the best client references and explains each choice; **80% R@1**, **90% R@5**, **0.85 MRR** on a 100-question internal evaluation.
- Built the platform's **evaluation harness** (retrieval metrics, LLM-reranker checks) and championed eval-driven development across A³, surfacing the need for AI governance to the **Chief AI Office**.

AI Engineer

Independent — R&D & Open Source

Feb. 2026 – June 2026

Paris, France

- Open-sourced **codibox** (PyPI), a Docker-sandboxed Python execution engine with namespace isolation, a drop-in tool for safe AI code execution in LangGraph.
- Designed a **multi-agent real estate platform** (LangChain deepagents, **6 subagents**, **15+ custom tools**) for valuation, ROI, and market research, cutting research time from **~3 hours to 15 minutes**.
- Built a **GraphRAG platform** on Neo4j with hybrid vector/full-text retrieval over a large financial document corpus for LLM due diligence, compliance, and risk Q&A.

AI Engineer · (Fixed-term contract)

Impactera — Financial Consulting (ESG & data intelligence)

Feb. 2025 – Feb. 2026

Paris, France (Remote)

- Architected a supervisor-based **multi-agent platform** (**LangGraph**, 6 subagents, FastAPI, Docker) spanning document RAG, live API/NL-to-SQL access, web research, and report generation, cutting report generation from **half a day to 30 minutes**.
- Built a **self-correcting RAG pipeline** (Qdrant) with semantic chunking, relevance grading, hallucination checks, and source attribution for **audit-ready** answers, auto-fallback on failed validation; shaped by CEO and finance-consultant needs.
- Engineered **multi-LLM routing** (OpenAI, Anthropic, Gemini, Groq, Bedrock) with automatic fallback and async checkpointing, removing single-provider dependency and rate-limit-driven downtime.
- Built a nested **API-orchestration agent** that constructs tool-calling agents at runtime from **OpenAPI specs** (Tagetik CPM, client systems) with OAuth-secured access.

AI Engineer · (Fixed-term contract)

Tetricks — AI Solutions for Hospitality (École Polytechnique-backed)

June 2023 – June 2024

Remote

- Led Vision-LLM integration for automated hotel room description generation, **deployed on GCP**, working with the CEO and Dev team; **secured Bpifrance innovation funding**.
- Engineered a **vLLM serving stack** with AWQ/GPTQ quantization and dynamic batching, achieving **3× throughput** and **47% cost reduction** vs. baseline serving.
- Fine-tuned **Llama 2** on hospitality domain data for guest intent detection across **12+ intent categories**, **deployed into production** guest-messaging workflows.
- Built a **multimodal RAG chatbot** on GCP (Vertex AI + Cloud Run) with a Pydantic-enforced ingestion pipeline and hybrid vector/lexical retrieval with reranking.

ML Research Assistant

Ibn Tofail University — Research Lab

Jan. 2022 – June 2023

Kenitra, Morocco

- Developed a **medical RAG chatbot** (Meditron 7B, Qdrant, PubMedBERT) for evidence-grounded clinical Q&A, reducing hallucination risk via source-cited retrieval.
- Built **North**, a dual-stack North African dialect classifier (scikit-learn + fine-tuned BERT) across Arabic and Arabizi Latin script, achieving **94.5%** test accuracy.
- Benchmarked state-of-the-art computer vision architectures on FER2013, achieving **90%** average precision and **20%** generalization improvement through targeted augmentation and regularization.

Leadership & Community

Founder & President

1337 AI — Educational Organisation

Oct. 2019 – Jan. 2022

Ben Guerir, Morocco

- Founded and scaled a **300-member** AI engineering community from scratch; ran **2 bootcamps** and **6 workshops** on Python, ML, and generative AI for student engineers.
- Co-mentor at the **ThinkAI 2023** and **HackAI 2024–25** hackathons, alongside Hugging Face and Mistral mentors.

Technical Skills

Agentic AI

LangGraph

LangChain

CrewAI

Google ADK

DSPy

MCP

Agentic RAG

RAG & Document Intelligence

RAG

GraphRAG

Hybrid Search

Reranking

Qdrant

Neo4j

pgvector

Document AI / OCR

VLMs

Multimodal Retrieval

LLM Engineering

vLLM

AWQ/GPTQ

LoRA/QLoRA

Inference Optimization

Multi-LLM Routing

Structured Generation

Fine-Tuning

Evaluation & Observability

RAGAS

LangSmith

Arize Phoenix

LLM-as-Judge

Agent Evaluation

OpenTelemetry

Prometheus

Grafana

AI Systems, Cloud & MLOps

Azure

GCP

AWS

Docker

Kubernetes

Terraform

CI/CD

Azure DevOps

MLflow

GitOps

Security & Governance

LLM Red-Teaming

Guardrails

Prompt-Injection Defense

Programming

Python

FastAPI

Pydantic

SQL

C++

CUDA

PyTorch

Education

IT Architecture Expert — Data Architecture

RNCP Level 7 · École 42 Paris, Paris, France

Master's in Artificial Intelligence

Ibn Tofail University, Kénitra, Morocco

BSc (Licence) in Computer Science

Ibn Tofail University, Kénitra, Morocco

Digital Technology Architect — Common Core

École 1337, UM6P, Ben Guerir, Morocco

Academic Experience

Systems & Hardware-Accelerated AI · École 42 Paris

Oct. 2024 – Feb. 2025

- Built a **CUDA-accelerated C/C++ linear algebra engine** (matrix ops, inverses, determinants).
- Inception of Things**: **K3s/K3D** clusters with **ArgoCD** GitOps and **GitLab** CI/CD.

Certifications

GCP AI Infrastructure

GPU Programming

GenAI Winter School

Red Teaming LLM Apps

LangGraph

Advanced RAG

Deep Learning Specialization

Google Cloud 2026

Johns Hopkins 2026

École Polytechnique 2024

DeepLearning.AI 2025

LangChain 2025

DeepLearning.AI 2024

DL.AI 2020

Open Source & Projects

🔗 **codibox** — sandboxed Python exec for agents · PyPI

🔗 **Real Estate Deep Agents** — multi-agent platform

🔗 **AgenticDataAnalyst** — NL-to-Python data analysis

Languages

Arabic (native)

French (B2)

English (C2)